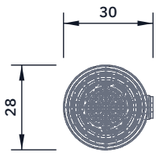
CONCEPT, NOT FOR FABRICATION

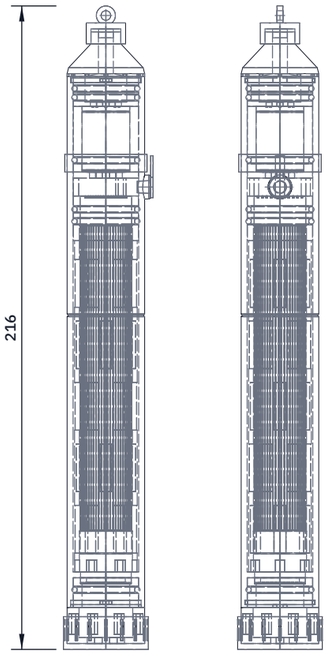
Helpful Water A: filter straw · github.com/BoujeeEnjinia1701/helpful-water

REV	DESCRIPTION	DATE	BY
P1	Preliminary GA from the TRL 3 model (HWA-CAL-001)	2026-11-11	AC
P2	Flow at 20 C and seal tolerances after the independent STEM review	2026-11-11	AC



TOP VIEW

Scale 1:2; looking down (along -Z)



FRONT VIEW

Scale 1:2; looking at the front (along +Y)

RIGHT VIEW

Scale 1:2; looking at the right side (along -X)



ISOMETRIC VIEW

Not to scale; lanyard cord not shown; seen from the front right and above, about 30 deg elevation

KEY DIMENSIONS (MM) AND DATA

Body 26 OD x 22 bore x 2 wall; 205 long without caps
Intake cap 0 to 22 (bayonet); seat ring 22 to 24, bore 17
Module 24 to 150: potting 30 to 38 and 140 to 150
About 220 fibres 0.8/0.5, active 102; 0.056 m2
Clean header 150 to 156; backflush socket at 153 on +X
Carbon cartridge 156 to 180: bed 16 dia x 20 (4.0 mL)
Mouthpiece 180 to 205; tip 10 dia, bore 5
O-rings AS568-018 in 19.3 +-0.05 glands; bore 22.0 +-0.05
Dip line at 110: immerse no deeper
Flow at 10 kPa 110 to 179 mL/min (clean, estimate)
Mass about 66 g; parts USD 17.46
PRELIMINARY, NOT FOR FABRICATION

HELPFUL WATER A: FILTER STRAW

github.com/BoujeeEnjinia1701/helpful-water

Hollow-fibre filter straw with backflush (option A1): general arrangement, caps fitted

DRAWING NO.	REV	SHEET	SIZE
HWA-DWG-001	P2	1 of 1	ANSI B

SCALE	UNITS	PROJECTION	DATE	DRAWN
1:2	mm		2026-11-11	Amish Chadha

MATERIAL / NOTES

PP parts (printed for the prototype); 316L mesh; EPDM O-rings; bought fibre module; GAC. See bom/bom.csv

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